An Awesome Introduction to Web Development

Assiut University

JavaScript

```html
<!DOCTYPE html>
<html>
<body>

<h2>My First JavaScript</h2>

<button type="button"
onclick="document.getElementById('demo').innerHTML = Date()">
Click me to display Date and Time.</button>

<p id="demo"></p>

</body>
</html>
```

My First JavaScript

Click me to display Date and Time.

Sun Oct 22 2023 16:47:47 GMT+0300 (EEST)

Why Use JavaScript?

JavaScript enables web designer to do programming in the HTML

- Read elements from documents and write new elements and text into documents
- Manipulate or move text
- Create pop-up windows
- Perform mathematical calculations on data
- React to events, such as a user rolling over an image or clicking a button

Why Use JavaScript?

- Retrieve the current date and time from a user’s computer.
- Determine the user’s screen size, browser version, or screen resolution
- Perform actions based upon conditions such as alerting users if they enter the wrong information into a form or if they press a certain button.

Adding a Script to Your Pages

- like CSS rules, JavaScript can either be embedded in a page or placed in an external script file.
- But in order to work in the browser, the browser must support JavaScript *and* must have it enabled
- You add scripts to your page inside the <script> element. The type attribute on the opening <script> tag indicates what scripting language will be found inside the element.
- Other languages exist like VB script and perl not only javascript.

Where to Add a Script to Your Pages?

There are three places where you can put your JavaScript

1-In the <head> of a page: These scripts will be called when an event triggers them.

2-In the <body> section: These scripts will run as the page loads.

3- In an external file: You can also write JavaScript in external documents that have the file extension .js. you need to use the src attribute on the <script> element.

```
<script type="JavaScript" src="scripts/validation.js" />
```

Comments in JavaScript

```
<script type="text/javascript">
document.write("My first JavaScript") //to eof line comment goes here
</script>

/* This whole section is commented
out so it is not treated as a part of
the script. */
```

The <noscript> Element

- What I should do (as a browser) if I can’t run js
- Offers alternative content for users whose browsers do not support JavaScript.

JavaScript on page load

- If you ever need to execute JavaScript on page load and manipulate elements, you might still consider placing the script at the end of the <body> or using the defer attribute, as they ensure the DOM elements are available. But as it stands, your function is perfectly fine for user-triggered actions.

How to use defer

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DOM Manipulation Example</title>
    <script defer>
        function changeText() {
            document.getElementById("header").innerText = "Text changed!";
        }
    </script>
</head>
<body>
    <h1 id="header">Hello, World!</h1>
    <button onclick="changeText()">Click me to change text</button>
</body>
</html>
```

Document Object Model (DOM)

The document object model explains what properties of a document a script can retrieve and which ones it can alter;

It also defines some methods that can be called upon the document.

The document object model specifies how you can retrieve values users have entered into a form. Once you have retrieved these values, you can use JavaScript to ensure the user has entered an appropriate value for that form control.

Key Concepts of the DOM

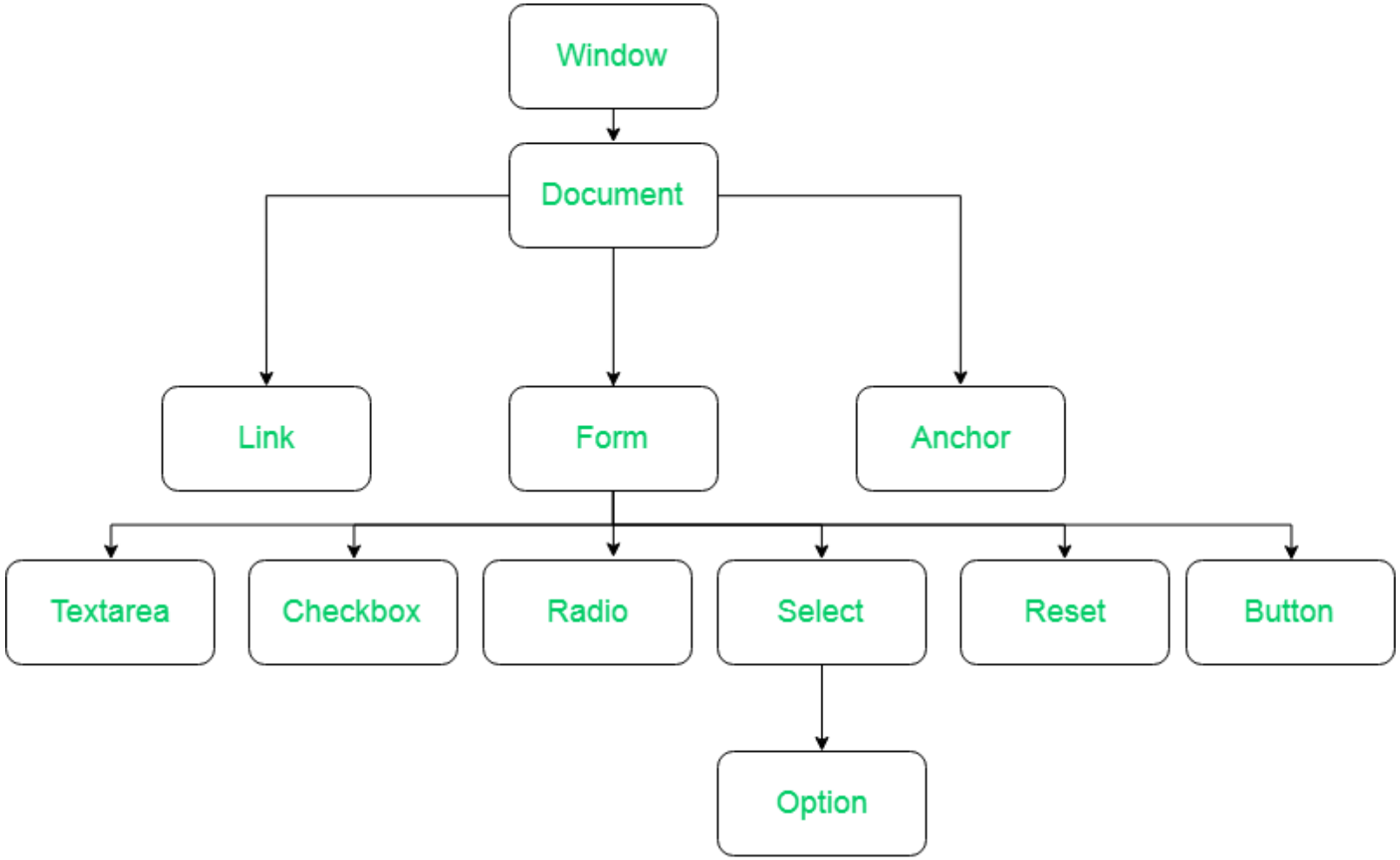
- **DOM Structure (Tree Structure)**
  - The DOM represents an HTML document as a tree of nodes. Each element, attribute, and piece of text in the HTML file becomes a "node" in this structure.
  - The root node of this tree is the <html> element, which has child nodes like <head>, <body>, etc., which in turn have their own child nod

```
<html>
  <body>
    <h1>Hello World</h1>
    <p>This is a paragraph.</p>
  </body>
</html>
```

```
html
├── head
└── body
    ├── h1
    └── p
```

DOM

- DOM Nodes and Elements
  - A node is any item in the DOM tree, including elements, text, and attributes.
  - An element node represents an HTML tag (like , <h1>, or <p>).
  - Text nodes are the text content inside elements.
  - Attribute nodes represent HTML attributes (such as id, class, or href).



DOM Figure

The Figure shows you how the elements of a page are made available in scripts as scriptable objects. The Document object represents the whole document and then each of the child objects represents a collection of similar tags within that document:

- The *forms* collection contains all of the <form> tags in the document.
- The *image* collection represents all of the images in a document.
- The *link* collection represents all of the hyperlinks within a page.
- The *anchor* collection represents all of the anchors in a document (<a> elements with a name or id attribute rather than an href attribute).
- The *area* collection represents all of the image maps that use an <area> element in the document.
- The *applet* collection represents all of the applets within a document

Accessing the DOM with JavaScript

JavaScript can interact with the DOM using various methods and properties provided by the document object. Here are some commonly used ones:

•Selecting Elements

- ❖ document.getElementById("id"): Selects an element with a specific ID.
- ❖ document.getElementsByClassName("class"): Selects elements with a specific class.
- ❖ document.getElementsByTagName("tag"): Selects elements by tag name.
- ❖ document.querySelector("selector"): Selects the first element that matches a CSS selector.
- ❖ document.querySelectorAll("selector"): Selects all elements that match a CSS selector.

Modifying Elements

- **element.innerHTML:**
  - Allows you to get or set the HTML content inside an element.
- **element.style.property:**
  - Allows you to set the CSS properties of an element, e.g., element.style.color = "red".
- **element.setAttribute("attribute", "value"):**
  - Sets an attribute of an element, e.g., element.setAttribute("class", "newClass").

Event Handling in the DOM

- he DOM allows JavaScript to handle events, such as clicks, key presses, and page loads:
  - Attach event listeners with addEventListener() to run specific code when a user interacts with an element.

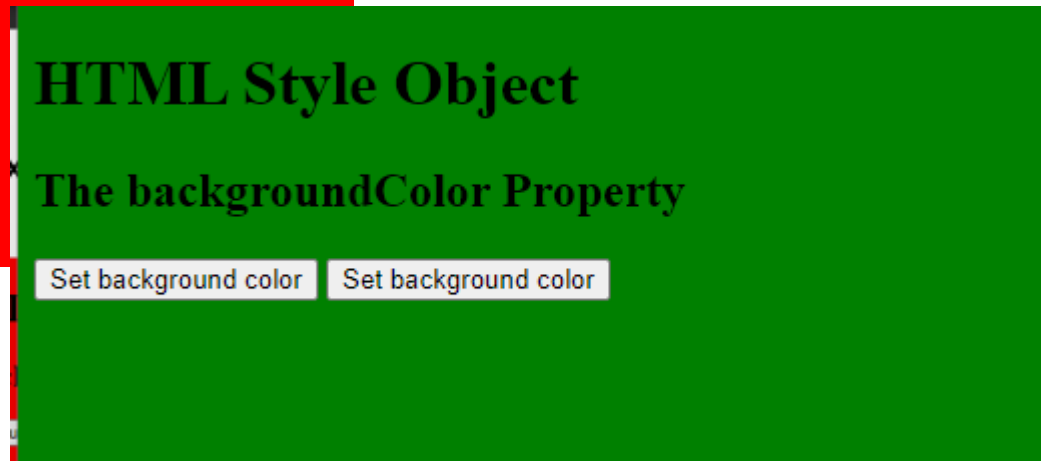
```
document.getElementById("myButton").addEventListener("click",
function() {    alert("Button clicked!");});
```

Example

```
<!DOCTYPE html>
<html>
<head>
<script>
function myFunction() {
document.body.style.backgroundColor = "red";
}
function myFunction2() {
document.body.style.backgroundColor = "green";
}
</script>
</head
<body>
<h1>HTML Style Object</h1>
<h2>The backgroundColor Property</h2>
<button type="button" onclick="myFunction()">Set background
color</button>
<button type="button" onclick="myFunction2()">Set background
color</button>
</body>
</html>
```

Example

HTML Style Object
The backgroundColor Property
Set background color
Set background color


HTML Style Object
The backgroundColor Property
Set background color
Set background color

Example (get Element by ID and apply Style by JS)

```
document.getElementById("myCSSid").innerHTML = "Paragraph contents changed!";

document.getElementById("myCSSid").src = "OlympiaLogo.jpg";

document.getElementById("myCSSid").style.color = 'red';
```

Example

The `classList` property is read-only, but you can use the methods listed below, to add, toggle or remove CSS classes from the list

```
<!DOCTYPE html>
<html>
<head>
<script>
function myFunction() {
  document.body.classList.toggle("mode");
}
</script>
<style>
.mode{
background-color:red;
color:white;
}
</style>
</head
<body>
<h1>HTML Style Object</h1>
<h2>The classList Property</h2>
<button type="button"
onclick="myFunction()">Toggle</button>

</body>
</html>
```

HTML Style Object
The classList Property
Toggle
HTML Style Object
The classList Property
Toggle

Potential Enhancements

```
.mode {
  background-color: red;
  color: white;
  transition: background-color 0.3s, color 0.3s;
}
```

- **CSS Transitions for Smoother Effect**
- Adding a transition effect can make the style change smoother

Back Button

```
<BUTTON onclick="history.back()">  GO BACK  </BUTTON>
```

An Example for further slides

```
<!DOCTYPE html>
<html>
<head>
</head>
<body>
<h1>User Registration</h1>
<form name="frmLogin" action="login.asp" method="post">
Username <input type="text" name="txtUsername" size="12" /> <br />
Password <input type="password" name="pwdPassword" size="12" /> <br />
<input type="submit" value="Log In" />
</form>
<p>If you are a new user <a href="register.asp">Register here</a> |
If you have lost your password you can <a href="lostPassword.asp">retrieve
your password here</a></p>
</body>
</html>
```

DOM Usage

The DOM would therefore make the content of the form available for use in the script as part of the forms collection and the links as part of the links collection.

```
document.links[0].href
```

There are four parts of this statement

1-The word document indicates I am accessing the document object.

2-The word links corresponds to the links collection.

3-The [0] indicates that I want the first link in the document.

4- You want to retrieve the href property for this link.

Alternative DOM Usage

An alternative approach is to use the names of elements to navigate a document. For an example, the following line requests the value of the password box:

```
document.frmLogin.pwdPassword.value
```

Again there are four parts to this statement:

1-The document comes first again as it is the top-level object.

2- Next you can see the name of the form frmLogin.

3- This is followed by the name of the form control pwdPassword.

4- Finally the property I am interested in is the value of the password box, and this property is called value

Objects, Methods, and Properties

An object model (such as the document object model) is made up of several objects, each of which can have *properties* and *methods*:

A property tells you something about an object.

A method performs an action.

Properties of the Document Object

Property	Description
document.anchors	Returns all <a> elements that have a name attribute
document.applets	Deprecated
document.baseURI	Returns the absolute base URI of the document
document.body	Returns the <body> element
document.cookie	Returns the document's cookie
document.doctype	Returns the document's doctype
document.documentElement	Returns the <html> element
document.documentMode	Returns the mode used by the browser
document.documentURI	Returns the URI of the document
document.domain	Returns the domain name of the document server
document.domConfig	Obsolete.
document.embeds	Returns all <embed> elements

Properties of the Document Object

document.forms	Returns all <form> elements
document.head	Returns the <head> element
document.images	Returns all <img> elements
document.implementation	Returns the DOM implementation
document.inputEncoding	Returns the document's encoding (character set)
document.lastModified	Returns the date and time the document was updated
document.links	Returns all <area> and <a> elements that have a href attribute
document.readyState	Returns the (loading) status of the document
document.referrer	Returns the URI of the referrer (the linking document)
document.scripts	Returns all <script> elements
document.strictErrorChecking	Returns if error checking is enforced
document.title	Returns the <title> element
document.URL	Returns the complete URL of the document

Methods of the Document Object

write(string**)** Allows you to add text or elements into a document

writeln(string**)** The same as *write()* but adds a new line at the end of the output

The Forms Collection

The forms collection holds references corresponding to each of the <form> elements in the page.

If the login form is the first form in the document and you want to access the action property

document.forms[0].action

Alternatively, you can directly access that form object using its name:

document.frmLogin.action

Properties of the Form Objects

action, length, method, name, target

Methods of the Form Objects

reset() Resets all form elements to default values

submit() Submits the form

Form Elements

When you access a form you usually want to access one or more of its elements.

Each form element has an **elements[]** collection object as a property

Example

```
<html>
<body>
<form id="myForm" action="/action_page.php">
First name: <input type="text" name="fname" value="Donald"><br>
Last name: <input type="text" name="lname" value="Duck"><br>
<input type="submit" value="Submit">
</form>
<p>Click the "Try it" button to display the number of elements in the form.</p>
<button onclick="myFunction()">Try it</button>
<p id="demo"></p>
<script>
function myFunction() {
var x = document.getElementById("myForm").elements.length;
document.getElementById("demo").innerHTML = "Found " + x + " elements in the form.";
}
</script>
</body>
</html>
```

First name: Donald

Last name: Duck

Submit

Click the "Try it" button to display the number of elements in the form.

Try it

Found 3 elements in the form.

What to do with the Elements?

Here are some of the things you might want to do with the elements in a form:

1- **Text fields:** Read data a user has entered or write new text to these elements.

2- Checkboxes and radio buttons: Test if they are checked and check or uncheck them.

3- **Buttons:** Disable them until a user has selected an option.

4- **Select boxes: see next slides**

1. Reading the Selected Option

- Access the current selection made by the user.
- Example:

- let selectedValue = document.getElementById("selectBox").value;
- console.log(selectedValue); // Logs the selected option's value

2. Setting a Default or Preselected Option

- Programmatically set a specific option as selected.
- Useful for setting a default choice or updating selection.

- Example:

- document.getElementById("selectBox").value = "option2";

.3Disabling or Enabling Options

- Limit user choices by disabling certain options.
- Useful for conditional selections.

- Example:

- document.getElementById("selectBox").options[2].disabled = true;

.4Adding or Removing Options

- Dynamically add or remove options in the dropdown.
- Useful in interactive forms with dependent options.

- Example to add an option:

- let newOption = new Option("New Option", "newOptionValue;("
- document.getElementById("selectBox").add(newOption;(

.5Resetting Selection

- Reset the dropdown to its initial state or a specific option.
- Example:
- document.getElementById("selectBox").selectedIndex = ;0

Summary

- Each interaction allows for dynamic dropdown management:
- -Reading the selected option
- -Setting default options
- -Enabling or disabling options
- -Adding or removing options
- -Resetting selection

- These help in building interactive, user-friendly forms!

Properties of Form Elements

- Checked Checkboxes and radio buttons Returns true when checked or false when not Read/write
- disabled All except hidden Returns true when disabled and user cannot interact with it (supported in IE 4 and Netscape 6 and later versions only)

Properties of Form Elements

- Form All elements Returns a reference to the form it is part of Read only
- length Select boxes Number of options in the <select> element
- name All elements Accesses the name attribute of the element (Read only)

Properties of Form Elements

- selectedIndex Select boxes Returns the index number of the currently selected item (Read/Write)
- type All Returns type of form control Read only
- value All Accesses the value attribute of

Three Types of Objects in JavaScript

1-W3C DOM objects:

like document object, browser object,..

2-Built-in Objects:

such as date object which deal with date and time , math object

3-Custom objects:

for advanced JavaScript ,you can write your own objects an example

Start Programming in JavaScript

Programming is the capability to make calculations in the HTML documents and it has many of the features and concepts in programming languages:

1- A variable is used to store data of different data types.

2-Operators: Arithmetic, Relational, Logic, Assignment, String.

3-Conditional statements

4-Loops

5-Functions

Variables

There are a few rules you must remember about variables in JavaScript:

- Variable names are case-sensitive.
- They must begin with a letter or the underscore character.
- Avoid giving two variables the same name within the same document as one might override the value of the other, creating an error.
- Try to use descriptive names for your variables.

Variable Declaration

Variables are declared by var keyword e.g

var userName = “Ahmed Aly"
userName = “’Mourad Aly“

```
var   x, y, z
 x=“ javaScript programming “
 y=1240
 z=1234.567
```

Lifetime of a Variable

Variables declared in a function are local to this function scope.

The same variable name can be used in different functions.

Variables declared outside functions are global variables and can be used by any function.

Arithmetic Operators

Symbol Description Example (x = 10) Result

- + Addition x+5 15
- - Subtraction x-2 8
- * Multiplication x*3 30
- / Division x/2 15
- % Modulus (division remainder) x%3
- ++ Increment (increments the variable by 1—this technique is often used in counters) if x = 10 then x++ → 11
- -- Decrement (decreases the variable by 1) x-- → 9

Assignment Operators (=)

Variable = variable or expression

total = total – profit

Shorthand

Symbol	Using Sh	Equi. Without SH
• +=	x+=y	x=x+y
• -=	x-=y	x=x-y
• *=	x*=y	x=x*y
• /=	x/=y	x=x/y
• %=	x%=y	x=x%y

Comparison Operators

Operator	Description	Example
• ==	is equal to	1==2 returns false 3==3 returns true
• !=	not equal to	1!=2 returns true 3!=3 returns false
• >	greater than	1>2 false
• >=	grater than or eq to	3>=2 true
• <	less than	2< 3 true
• <=	less than or equal to	3<=3 true

Logical Operator

Operator	Name	Desc Example (x=1 and y=2)
• &&	And	Allows you to check if both of two conditions are met (x < 2 && y > 1) Returns true
• ||	Or	Allows you to check if one of two conditions are met (x < 2 || y < 2) Returns true
• !	Not	Allows you to check if something is not the case ! (x > y) returns true

String Operator

```
firstName = "Bob"

lastName = "Stewart"

name = firstName + lastName
```

Functions

- A function is just like a method, except a method belongs to an object, and a function lives on its own at the top of a document—but both perform actions.

How to Define a Function

There are three parts to creating or defining a function:
1-Define a name for it.
2-Indicate any values that might be required as arguments.
3-Add statements.

```
function calculateArea(width, height)
 {
        area = width * height
        return area
}
```

How to Call a Function

```
<form name="frmArea" action="">
	Enter the width and height of your rectangle to calculate the size:<br />
	Width: <input type="text" name="txtWidth" size="5" /><br />
	Height: <input type="text" name="txtHeight" size="5" /><br />
	<input type="button" value="Calculate area"
	onclick="alert(calculateArea(document.frmArea.txtWidth.value,
	document.frmArea.txtHeight.value))" />
</form>
```

Conditional Statements

- if statements, which are used when you want the script to execute if a condition is true

Single selection

- if...else statements, which are used when you want to execute one set of code if a condition is true and another if it is false

Double selection

- switch statements, which are used when you want to select one block of code from many

Multi Selection

Conditional Statements Example

```
<script type="text/JavaScript">
        date = new Date();
        time = date.getHours();
        if (time < 12)
                document.write('Good Morning’);
        else
                document.write('Good Afternoon');
</script>
```

Conditional Statements Example

```
<script>
  let day = new Date().getDay(); // returns 0 (Sun) to 6 (Sat)
  let dayName;
  switch (day) {
    case 0:
      dayName = "Sunday";
      break;
    case 1:
      dayName = "Monday";
      break;
    case 2:
      dayName = "Tuesday";
      break;
    case 3:
      dayName = "Wednesday";
      break;
    case 4:
dayName = "Thursday";
      break;
    case 5:
      dayName = "Friday";
      break;
    case 6:
      dayName = "Saturday";
      break;
    default:
      dayName = "Unknown day";
  }

  document.getElementById("result").textContent = "Today is: " + dayName;
</script>
```

Conditional (or Ternary) Operator

A conditional operator (also known as the ternary operator) assigns a value to a variable based upon a condition:

variablename=(condition)?value1:value2

JS cont.

Digging deeper into JS

Looping

- A ***while*** loop runs the same block of code while or until a condition is true.

- A ***do while*** loop runs once before the condition is checked. If the condition is true, it will continue to run until the condition is false.

- A ***for*** loop runs the same block of code a specified number of times.

The For Loop

The `for` statement creates a loop with 3 optional expressions:

```
for (expression 1; expression 2; expression 3) {
  // code block to be executed
}
```

Expression 1 is executed (one time) before the execution of the code block.
Expression 2 defines the condition for executing the code block.
Expression 3 is executed (every time) after the code block has been executed.

Example

Instead of writing:

```
text += cars[0] + "<br>";
text += cars[1] + "<br>";
text += cars[2] + "<br>";
text += cars[3] + "<br>";
text += cars[4] + "<br>";
text += cars[5] + "<br>";
```

You can write:

```
for (let i = 0; i < cars.length; i++) {
  text += cars[i] + "<br>";
}
```

Example

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript For Loop</h2>
<p id="demo"></p>
<script>
 cars = ["BMW", "Volvo", "Saab", "Ford", "Fiat", "Audi"];
text = "";
for (let i = 0; i < cars.length; i++) {
 text += cars[i] + "<br>";
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

JavaScript For Loop

BMW2
Volvo
Saab
Ford
Fiat
Audi

Loop vars scope using var

```
var i = 5;

for (var i = 0; i < 10; i++) {
  // some code
}

// Here i is 10
```

Using Let

- Using let, the variable declared in the loop does not redeclare the variable outside the loop.

- When let is used to declare the i variable in a loop, the i variable will only be visible within the loop.

```
let i = 5;

for (let i = 0; i < 10; i++) {
  // some code
}

// Here i is 5
```

The While Loop

The `while` loop loops through a block of code as long as a specified condition is true.

Syntax

```
while (condition) {
  // code block to be executed
}
```

Example

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript While Loop</h2>
<p id="demo"></p>
<script>
let text = "";
let i = 0;
while (i < 10) {
text += "<br>The number is " + i;
i++;
}
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

JavaScript While Loop

The number is 0
The number is 1
The number is 2
The number is 3
The number is 4
The number is 5
The number is 6
The number is 7
The number is 8
The number is 9

The Do While Loop

- The do while loop is a variant of the while loop.
- This loop will execute the code block once, before checking if the condition is true, then it will repeat the loop as long as the condition is true.

Syntax

- do {
- // code block to be executed
- }
- while (condition);

Example

```
<!DOCTYPE html>
<html>
<body>
<h2>JavaScript Do While Loop</h2>
<p id="demo"></p>
<script>
let text = ""
let i = 0;
do {
text += "<br>The number is " + i;
i++;
}
while (i > 10);
document.getElementById("demo").innerHTML = text;
</script>
</body>
</html>
```

JavaScript Do While Loop

The number is 0

Handling Events

- Window Event Attributes
- Form Events
- Keyboard Events
- Mouse Events
- Clipboard Events

Window Event Attributes

onload		Fires after the page is finished loading

```
<!DOCTYPE html>
<html>
<head>
<script>
function myFunction() {
  alert("Page is loaded");
}
</script>
</head>

<body onload="myFunction()">
<h1>Hello World!</h1>
</body>

</html>
```

Form Events

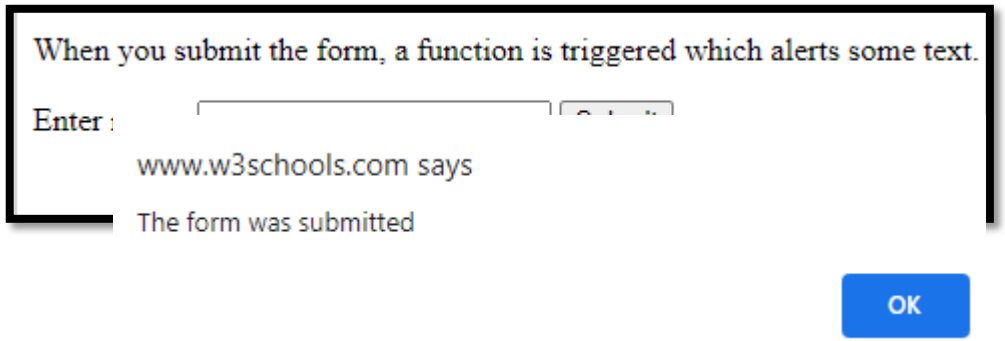
onsubmit		Fires when a form is submitted

```
<!DOCTYPE html>
<html>
<body>

<p>When you submit the form, a function is triggered which
alerts some text.</p>

<form action="/action_page.php" onsubmit="myFunction()">
  Enter name: <input type="text" name="fname">
  <input type="submit" value="Submit">
</form>

<script>
function myFunction() {
  alert("The form was submitted");
}
</script>
</body>
</html>
```

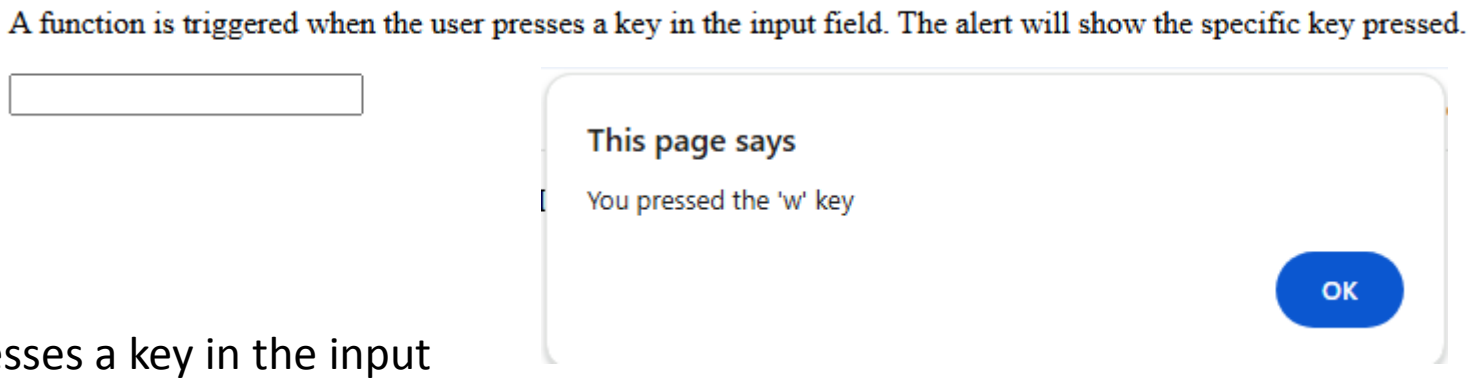


Keyboard Events

onkeydown		Fires when a user is pressing a key

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Show Pressed Key</title>
</head>
<body>
<p>A function is triggered when the user presses a key in the input field. The alert will show the specific key pressed.</p>

<input type="text" onkeydown="myFunction(event)">
<script>
function myFunction(event) {
  alert("You pressed the '" + event.key + "' key");
}
</script>
</body>
</html>
```



Mouse Events

ondblclick		Fires on a mouse double-click on the element

```
<!DOCTYPE html>
<html>
<body>

<button ondblclick="myFunction()">Double-click me</button>

<p id="demo"></p>

<p>A function is triggered when the button is double-clicked.
The function outputs some text in a p element with
id="demo".</p>

<script>
function myFunction() {
  document.getElementById("demo").innerHTML = "Hello
World";
}
</script>
</body>
</html>
```

Double-click me

A function is triggered when the button is double-clicked. The function outputs some text in a p element with id="demo".

Double-click me

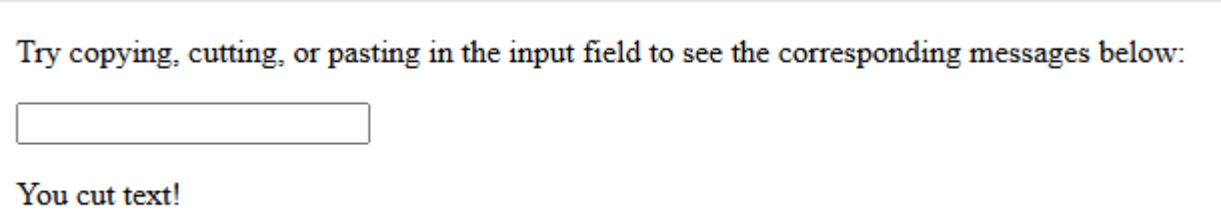
Hello World

A function is triggered when the button is double-clicked. The function outputs some text in a p element with id="demo".

Clipboard Events

Attribute	Description
oncopy	Fires when the user copies the content of an element
oncut	Fires when the user cuts the content of an element
onpaste	Fires when the user pastes some content in an element

```
<!DOCTYPE html>
<html lang="en">
<head>
<title>Copy, Cut, and Paste Events</title>
</head>
<body>
<p>Try copying, cutting, or pasting in the input field to see the corresponding messages below:</p>
<input type="text"
      oncopy="myFunction('copy')"       oncut="myFunction('cut’)”      onpaste="myFunction('paste')"
      value="Try to copy, cut, or paste this text">
<p id="demo"></p>
<script>
function myFunction(action) {   let message = "";
  switch(action) {
    case 'copy’:       message = "You copied text!";
      break;
    case 'cut’:       message = "You cut text!";
      break;
    case 'paste’:       message = "You pasted text!";
      break;   }
  document.getElementById("demo").innerHTML = message; } </script>  </body> </html>
```



Drag Events

Attribute	Value	Description
ondrag	*script*	Script to be run when an element is dragged
ondragend	*script*	Script to be run at the end of a drag operation
ondragenter	*script*	Script to be run when an element has been dragged to a valid drop target
ondragleave	*script*	Script to be run when an element leaves a valid drop target
ondragover	*script*	Script to be run when an element is being dragged over a valid drop target
ondragstart	*script*	Script to be run at the start of a drag operation
ondrop	*script*	Script to be run when dragged element is being dropped
onscroll	*script*	Script to be run when an element's scrollbar is being scrolled

Extra Various Events

onload, onunload, onclick, ondblclick,

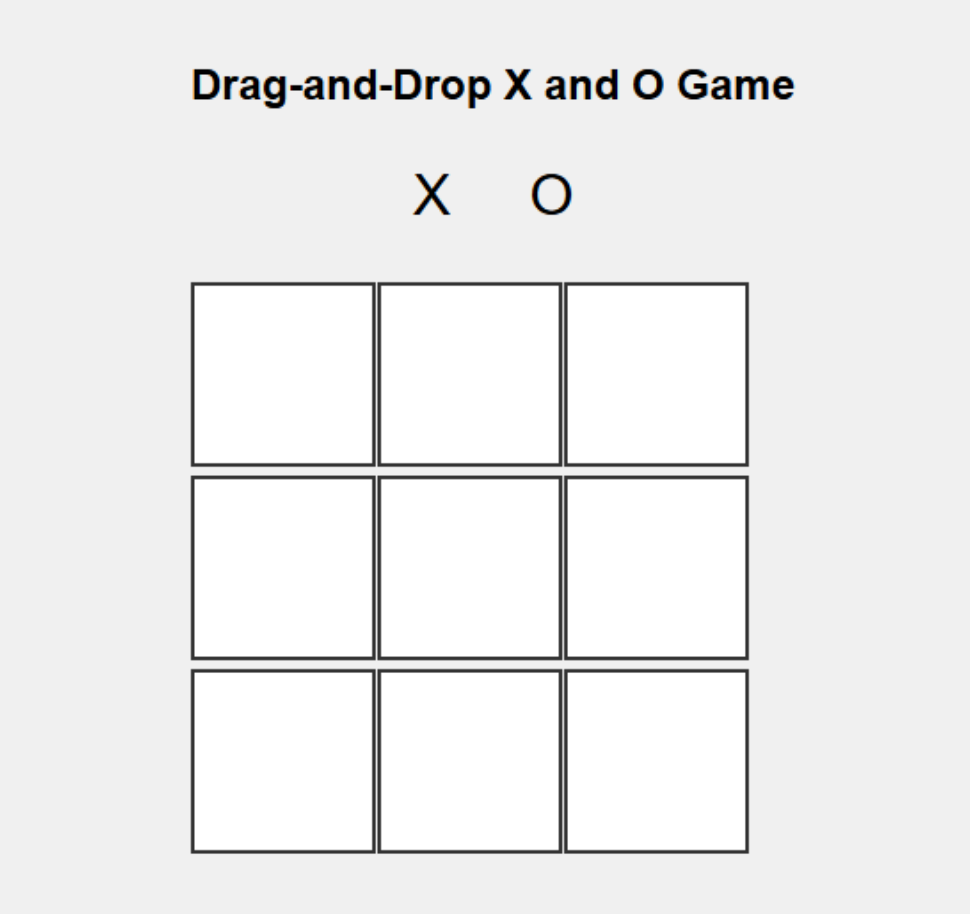
onmousedown, onmouseup, onmouseover,

onmousemove, onmouseout, onkeypress,

onkeydown, onkeyup, onfocus, onblur,

onsubmit, onreset, onselect, onchange

Drag and Drop XO: Try at Home



Objects

Built-in Objects

String

- The string object allows you to deal with strings of text.
- Before you can use a built-in object you need to create an instance of that object.
- You create an instance of the string object by assigning it to a variable like so:

```
myString = new String('Here is some big text')
```

String Properties and Methods

charAt()	Returns the character at a specified index (position)
length	Returns the length of a string
replace()	Searches a string for a pattern, and returns a string where the first match is replaced
split()	Splits a string into an array of substrings
trim()	Returns a string with removed whitespaces

String HTML Wrapper Methods

link()	Displays a string as a hyperlink

```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript String Methods</h1>
<h2>The link() Method</h2>
<p id="demo1"></p>
<p>The link() method is deprecated in JavaScript.</p>
<p>Use an a tag instead:</p>
<p id="demo2"></p>
<script>
let text = "Free Web Building Tutorials!";
let result = text.link("https://www.w3schools.com");
document.getElementById("demo1").innerHTML = result;

result = "<a href='https://www.w3schools.com'>" + text +
"</a>";
document.getElementById("demo2").innerHTML = result;
</script>

</body>
</html>
```

JavaScript String Methods

The link() Method

Free Web Building Tutorials!

The link() method is deprecated in JavaScript.

Use an a tag instead:

Free Web Building Tutorials!

Date

The ***date*** object helps you work with dates and times. You create a new date object using the date constructor like so:

```
new Date()
```

You can create a date object set to a specific date or time, in which case you need to pass it one of four parameters:

- **milliseconds:** This value should be the number of milliseconds since 01/01/1970.
- **dateString:** Can be any date in a format recognized by the parse() method.
- **yr_num, mo_num, day_num:** Represents year, month, and day.
- **yr_num, mo_num, day_num, hr_num, min_num, seconds_num:** Represents the hours, days, minutes, seconds, and milliseconds.

Date Examples

- **The first uses milliseconds and will read Thu Nov 27 05:33:20 UTC 1975:**

```
var birthDate = new Date(8298400000)
document.write(birthDate)
```

- **The second uses a dateString, and will read Wed Apr 16 00:00:00 UTC+0100 1975:**

```
var birthDate = new Date("April 16, 1975")
document.write(birthDate)
```

- **The third uses yr_num, mo_num, and day_num, and will read Mon May 12 00:00:00 UTC+0100 1975:**

```
var birthDate = new Date(1975, 4, 28)
document.write(birthDate)
```

Date Methods

getDate()	Returns the day of the month (from 1-31)
getDay()	Returns the day of the week (from 0-6)
getFullYear()	Returns the year
getHours()	Returns the hour (from 0-23)
getMilliseconds()	Returns the milliseconds (from 0-999)
getMinutes()	Returns the minutes (from 0-59)
getMonth()	Returns the month (from 0-11)
getSeconds()	Returns the seconds (from 0-59)

Math **Properties**

The math object helps in working with numbers

Property	**Purpose**
• E	Returns the base of a natural logarithm
• LN2	Returns the natural logarithm of 2
• LN10	Returns the natural logarithm of 10
• LOG2E	Returns the base-2 logarithm of E
• LOG10E	Returns the base-10 logarithm of E
• PI	Returns pi
• SQRT1_2	Returns 1 divided by the square root of 2
• SQRT2	Returns the square root of 2

Example

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Math Constants</h2>

<p id="demo"></p>

<script>
document.getElementById("demo").innerHTML =
"<p><b>Math.E:</b> " + Math.E + "</p>" +
"<p><b>Math.PI:</b> " + Math.PI + "</p>" +
"<p><b>Math.SQRT2:</b> " + Math.SQRT2 + "</p>" +
"<p><b>Math.SQRT1_2:</b> " + Math.SQRT1_2 + "</p>" +
"<p><b>Math.LN2:</b> " + Math.LN2 + "</p>" +
"<p><b>Math.LN10:</b> " + Math.LN10 + "</p>" +
"<p><b>Math.LOG2E:</b> " + Math.LOG2E + "</p>" +
"<p><b>Math.Log10E:</b> " + Math.LOG10E + "</p>";
</script>

</body>
</html>
```

JavaScript Math Constants

Math.E: 2.718281828459045

Math.PI: 3.141592653589793

Math.SQRT2: 1.4142135623730951

Math.SQRT1_2: 0.7071067811865476

Math.LN2: 0.6931471805599453

Math.LN10: 2.302585092994046

Math.LOG2E: 1.4426950408889634

Math.Log10E: 0.4342944819032518

Math Methods

Method	Purpose
• abs(x)	Returns the absolute value of *x*
• acos(x)	Returns the arccosine of *x*
• asin(x)	Returns the arcsine of *x*
• atan(x)	Returns the arctangent of *x*
• atan2(y,x)	Returns the angle from the x-axis to a point
• ceil(x)	Returns the nearest integer greater than or equal to*x*
• cos(x)	Returns the cosine of *x*
• exp(x)	Returns the value of E raised to the power of *x*

Math Methods cont.

Method	Purpose
• floor(x)	Returns the nearest integer less than or equal to *x*
• log(x)	Returns the natural log of *x*
• max(x,y)	Returns the number with the highest value of *x* and *y*
• min(x,y)	Returns the number with the lowest value of *x* and *y*
• pow(x,y)	Returns the value of the number *x* raised to the power of *y*

Math Methods cont.

Method	Purpose
• random()	Returns a random number between 0 and 1
• round(x)	Rounds *x* to the nearest integer
• sin(x)	Returns the sine of *x*
• sqrt(x)	Returns the square root of *x*
• tan(x)	Returns the tangent of *x*

Array

An *array* is like special variable

- instruments = new Array("guitar", "drums", "piano")
- instruments = new Array(3)

Window

Every browser window and frame has a corresponding Window object that is created with every instance of a <body> or <frameset> element.

Window Properties List

1. Window Width: **`window.innerWidth`** — The width of the window's viewport.
2. Window Height: **`window.innerHeight`** — The height of the window's viewport.
3. Screen Width: **`window.screen.width`** — The total width of the user's screen.
4. Screen Height: **`window.screen.height`** — The total height of the user's screen.
5. URL: **`window.location.href`** — The full URL of the current page.
6. Protocol: **`window.location.protocol`** — The protocol used in the URL (e.g., http: or https:).

Window Properties List

- Host: `window.location.host` — The hostname and port number of the current URL.
- Pathname: `window.location.pathname` — The path of the URL after the host.
- History Length: `window.history.length` — The number of entries in the browser's history stack.
- User Agent: `window.navigator.userAgent` — A string identifying the browser and operating system.
- Window Location: `window.location.toString()` — The full location object as a string.

Window Properties List

- Window Name: **`window.name`** — The name of the current window.
- Window Outer Width: **`window.outerWidth`** — The width of the entire browser window, including the interface.
- Window Outer Height: **`window.outerHeight`** — The height of the entire browser window, including the interface.
- Online Status: **`navigator.onLine`** — Indicates whether the browser is online or offline.
- Browser Language: **`navigator.language`** — The preferred language setting of the browser.

```html
<!DOCTYPE html>
<html lang="en">
<head>
<title>Window Properties Example</title>
</head>
<body>
<h2>JavaScript Window Properties</h2>
<div id="properties"></div>
<script>
  // Window properties to display
  const windowProperties = [
    { name: "Window Width", value: window.innerWidth },
    { name: "Window Height", value: window.innerHeight },
    { name: "Online Status", value: navigator.onLine ? "Online" : "Offline" }, // Network status
    { name: "Browser Language", value: navigator.language || navigator.userLanguage } // Browser language
  ];
  // Build HTML using a for loop and display it
  let html = "";
  for (let prop of windowProperties) {
    html += `<div><strong>${prop.name}:</strong> ${prop.value}</div>`;
  }
  document.getElementById("properties").innerHTML = html;
</script>
</body>
</html>
```

JavaScript Window Properties

Window Width: 1366
Window Height: 599
Screen Width: 1366
Screen Height: 768
URL: file:///C:/Users/AUC/Desktop/abc.html
Protocol: file:
Host:
Pathname: /C:/Users/AUC/Desktop/abc.html
History Length: 1
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/130.0.0.0 Safari/537.36
Window Location: file:///C:/Users/AUC/Desktop/abc.html
Window Name:
Window Outer Width: 1366
Window Outer Height: 720
Online Status: Online
Browser Language: en-US

Methods

Method	Purpose
• alert()	Displays an alert box containing a message and an OK button.
• back()	Same effect as the browser's Back button.
• close()	Closes the current window or another window if a reference to another window is supplied.

Tips for writing JavaScript

1-Has Someone Already Written This Script?

- http://www.HotScripts.com
- http://www.JavaScriptKit.com
- http://www.JavaScriptSource.com
- ***Reusable Functions***

Form Validation

Forms are usually validated using the onsubmit event handler, which triggers a validation function stored in the head of the document, so the values are checked when the user presses the Submit button.

- You have probably seen forms on Web sites that first ask you to enter a password, and then to re-enter it to make sure you did not mistype something.
- In such a form you might want to check a few things:
- ❑ That the username is of a minimum length
- ❑ That the password is of a minimum length
- ❑ That the two passwords match

Form Validation Example

```
function validate(form) {
        var returnValue = true;
        var username = frmRegister.txtUserName.value;
        var password1 = frmRegister.txtPassword.value;
        var password2 = frmRegister.txtPassword2.value;
        if(username.length < 6) {
                returnValue = false;
                alert("Your username must be at least\n6 characters long.\nPlease try again.");
                frmRegister.txtUserName.focus();
        }
        if (password1.length < 6) {
                returnValue = false;
                alert("Your password must be at least\n6 characters long.\nPlease try again.");
```

Form Validation Example

```
                frmRegister.txtPassword.value = "";
                frmRegister.txtPassword2.value = "";
                frmRegister.txtPassword.focus();
        }
        if (password1.value != password2.value) {
                returnValue = false;
                alter("Your password entries did not match.\nPlease try again.");
                frmRegister.txtPassword.value = "";
                frmRegister.txtPassword2.value = "";
                frm Register.txtPassword.focus();
        }
        return returnValue;
}
```

Text Control
Username:
testuser
Password:
Please confirm your password:
Log in
Internet Explorer Script Alert
Your password must be at least
6 characters long.
Please try again.
OK

Required Text Fields

Often you will want to ensure that a user has entered some value into a text field.

Use a loop to go through all of the required elements using a for loop, and if any of them are empty, return an error.

When you use this technique you need to have a class attribute that has a value of required on each form element that is required so the loop can tell if the text input must have a value, and you must have a name attribute whose value matches the label for the element (because this will be used in any error message).

Example

Here is an example of what a text input should looks like with the name and class attributes:

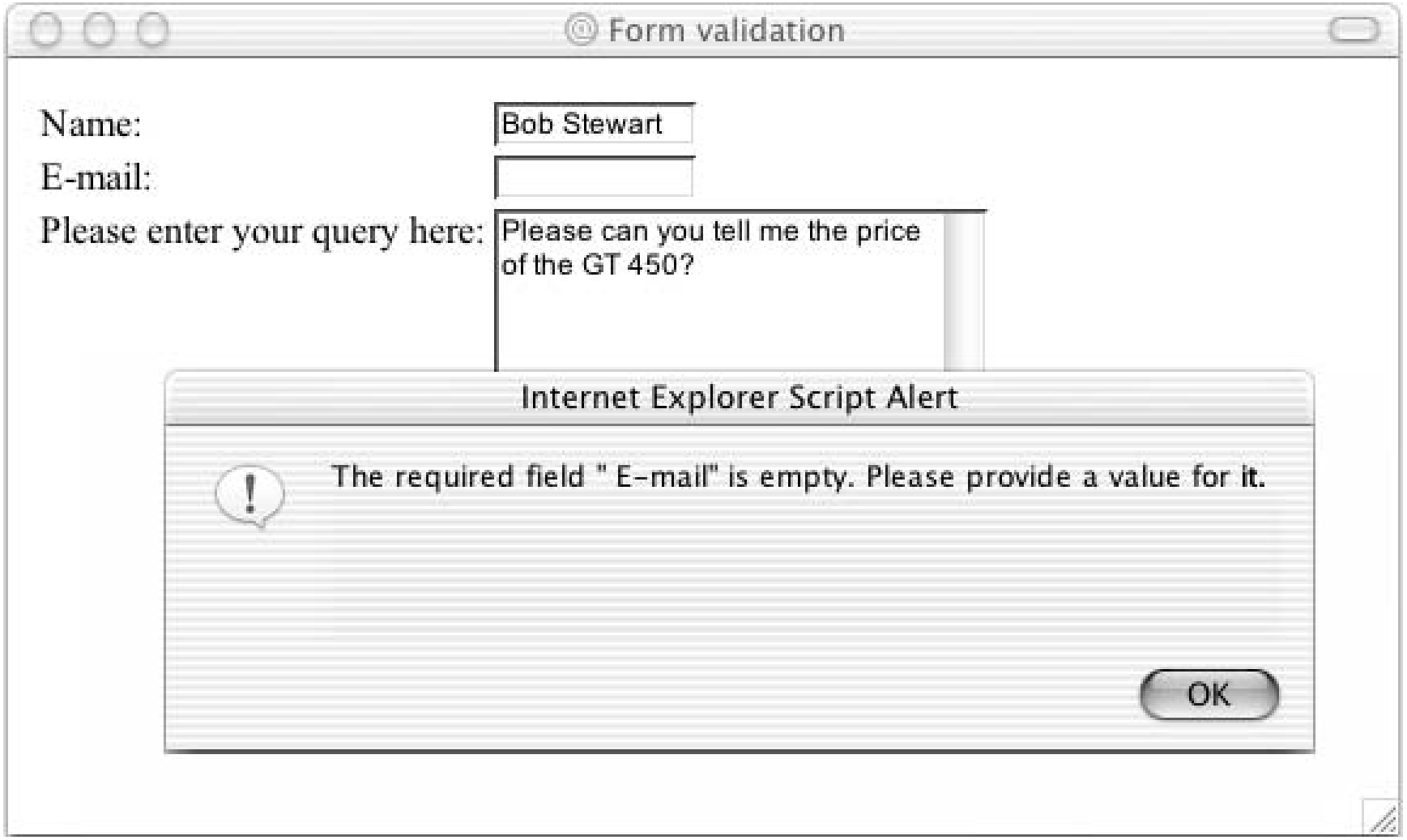
```
<input type="text" name="Username" size="5" class="required" />
```

Example

```
function validate(form) {
        var returnValue = true;
        var formElements = form.elements;
        for (var i=0; i<formElements.length; i++) {
                currentElement = formElements[i];
                if (currentElement.value=="" && currentElement.className=="required") {
                        alert("The rquired field \""+currentElement.name +"\" is empty. Please
                        provide a value for it.");
                        currentElement.focus();
                        returnValue = false;
                        break;
                }
                return returnValue;}}
```

```
<form name="frmEnquiry" method="post" action="register.asp” onsubmit="return validate(this);">
        <table>
        <tr>
                <td>Name:</td>
                <td><input type="text" class="required" name="Name" size="12" /></td>
        </tr>
        <tr>
                <td>E-mail: </td>
                <td><input type="text" class="required" name="Email" size="12" /></td>
        </tr>
```

```
        <tr>
                <td valign="top">Please enter your query here:</td>
                <td><textarea rows="8" class="required" cols="30" name="Query">
                </textarea></td> Creating a JavaScript Library
        </tr>
        <tr>
                <td> </td>
                <td><input type="submit" value="Submit your query" /></td>
        </tr>
</table>
</form>
```


Form validation
Name:
Bob Stewart
E-mail:
Please enter your query here:
Please can you tell me the price of the GT 450?
Internet Explorer Script Alert
The required field " E-mail" is empty. Please provide a value for it.
OK

Select Box & Radio

Select Box Options

- If you want to check whether a user has selected one of the items from a select box you need to use the selectedIndex property of the select object that represents the select box.

Radio Buttons

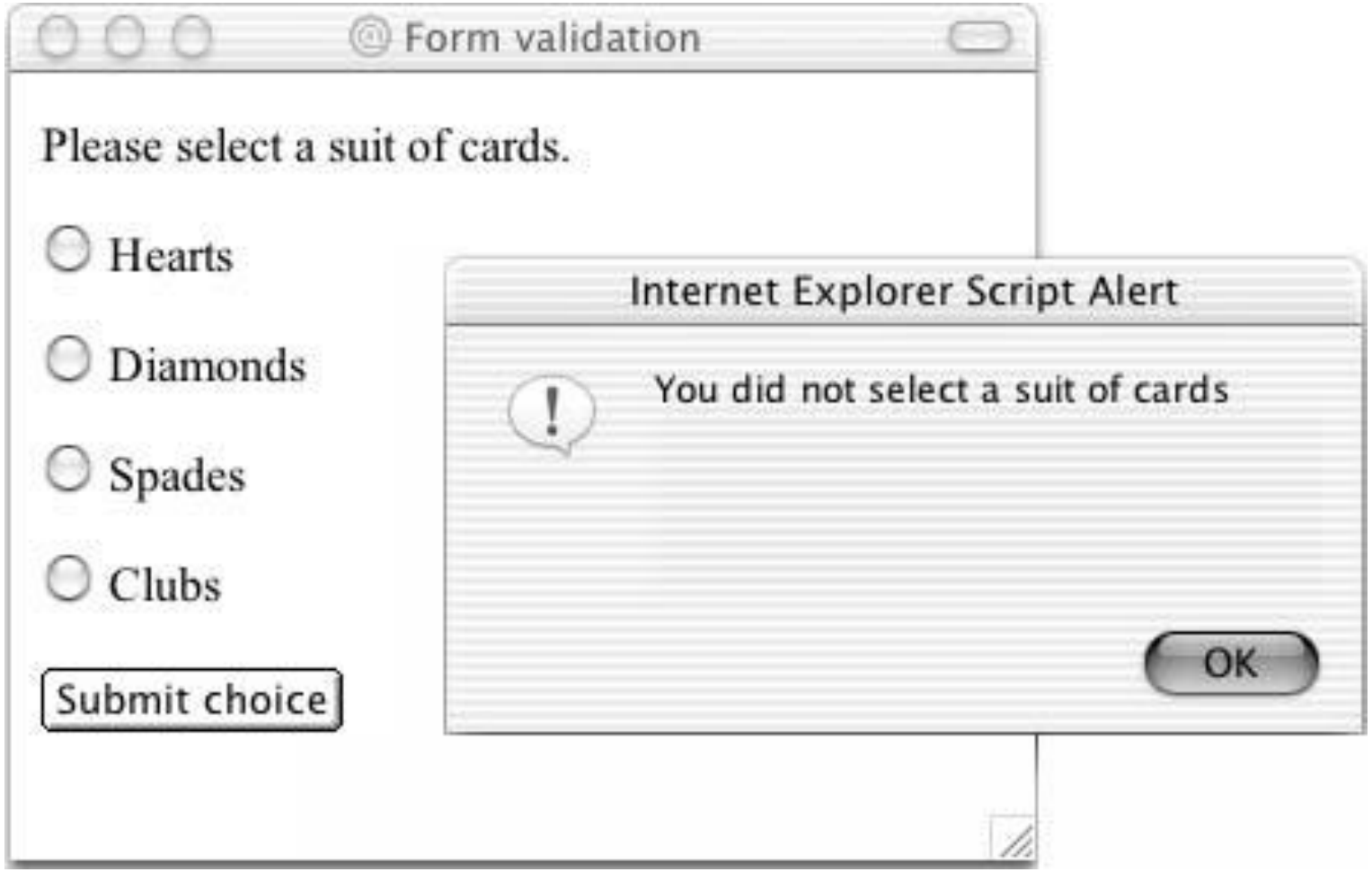
- If you want to ensure that a radio button has been selected you can either preselect one of the radio button values or you can loop through the RadioButton object’s checked properties to see if one has been selected.

An Example

```
<form name="frmCards" action="cards.asp" method="post” onsubmit="return validateForm(this)" >
        <p>Please select a suit of cards.</p>
        <p><input type="radio" name="radSuit" value="hearts" /> Hearts </p>
        <p><input type="radio" name="radSuit" value="diamonds" /> Diamonds </p>
        <p><input type="radio" name="radSuit" value="spades" /> Spades </p>
        <p><input type="radio" name="radSuit" value="clubs" /> Clubs </p>
        <p><input type="submit" value="Submit choice" /></p>
</form>
```

JS

```
function validate(form) {
- var radioButtons = form.radSuit;
- var radioChosen = false;
- for (var i=0; i<radioButtons.length; i++) {
  - if (radioButtons[i].checked)
  - {
    - radioChosen=true;
    - returnValue=true;
  - }
- }
- if (radioChosen == false) {
  - returnValue = false;
  - alert("You did not select a suit of cards");
- }
- return returnValue;
}
```


Form validation
Please select a suit of cards.
Hearts
Diamonds
Spades
Clubs
Submit choice
Internet Explorer Script Alert
You did not select a suit of cards
OK

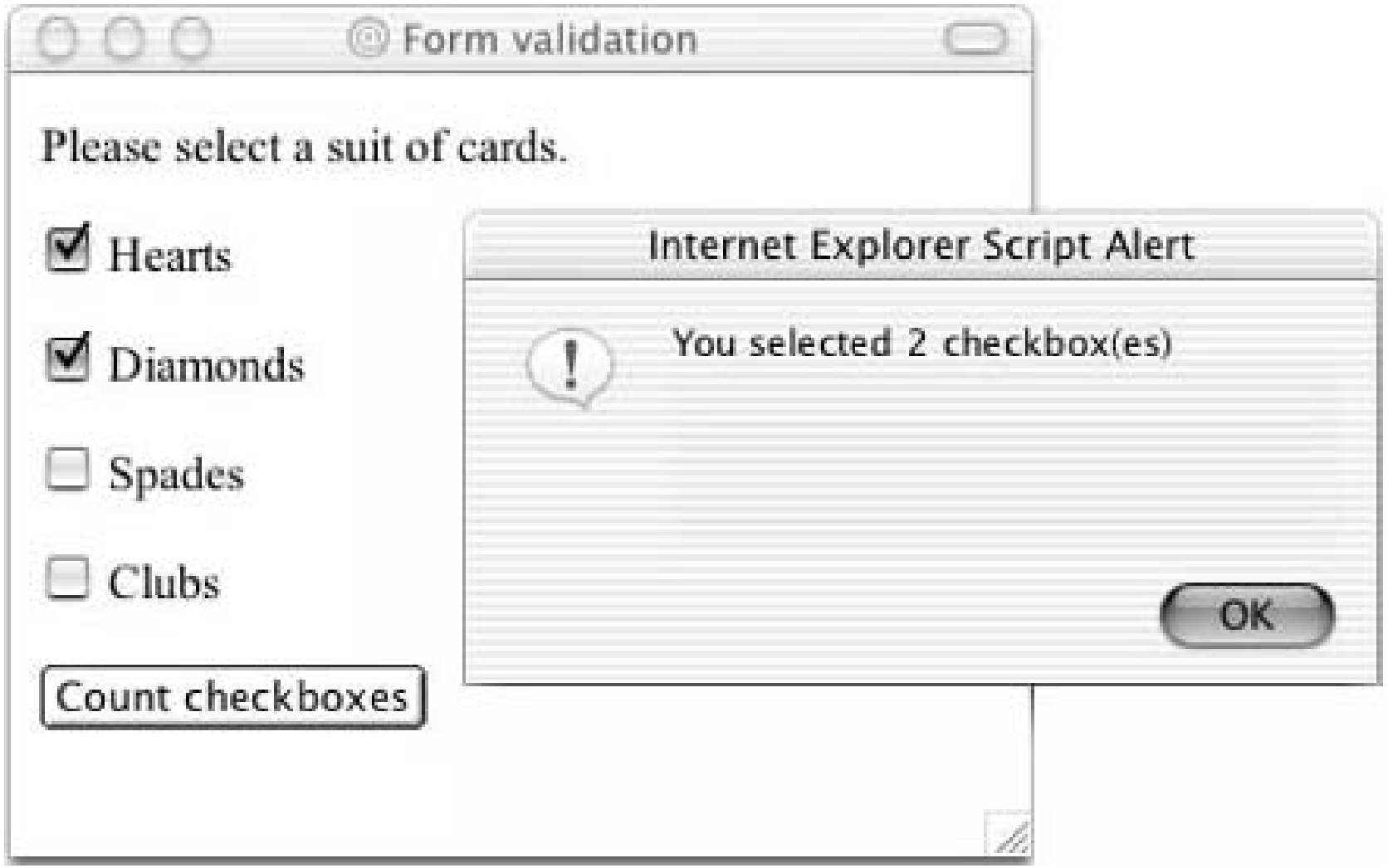
Validating Check Boxes

Checkboxes

```
<form name="frmCards" action="cards.asp" method="post">
        <p>Please select one or more suits of cards.</p>
        <p><input type="checkbox" name="chkSuit" value="hearts" /> Hearts </p>
        <p><input type="checkbox" name="chkSuit" value="diamonds" />
        Diamonds </p>
        <p><input type="checkbox" name="chkSuit" value="spades" /> Spades
        </p>
        <p><input type="checkbox" name="chkSuit" value="clubs" /> Clubs </p>
        <p><input type="button" value="Count checkboxes”
        onclick="countCheckboxes(frmCards.chkSuit)" /></p>
</form>
```

JS

```
function countCheckboxes(field) {
        var intCount = 0
        for (var i = 0; i < field.length; i++) {
                if (field[i].checked)
                        intCount++;
         }
        alert("You selected " + intCount + " checkbox(es)");
}
```


Form validation
Please select a suit of cards.
Hearts
Diamonds
Spades
Clubs
Count checkboxes
Internet Explorer Script Alert
You selected 2 checkbox(es)
OK

Example

Disabling a Submit Button Until a Checkbox Has Been Selected

If you want to ensure that a checkbox has been selected—for example, if you want a user to agree to certain terms and conditions

```
• <!DOCTYPE html>
• <html lang="en">
• <head>
•   <meta charset="UTF-8">
•   <title>Checkbox Agreement Example</title>
•   <script>
•     function checkCheckBox(myForm) {
•       if (!myForm.agree.checked) {
•         alert('You must agree to terms and conditions if you want to download this product.');
•         return false;
•       } else {
•         return true;
•       }     }
•   </script>
• </head>
```

```
<body>
<form onsubmit="return checkCheckBox(this);">
  <label>
    <input type="checkbox" name="agree">
    I agree to the terms and conditions
  </label>
  <br><br>
  <input type="submit" id="submitButton" value="Download" >
</form>
</body>
</html>
```

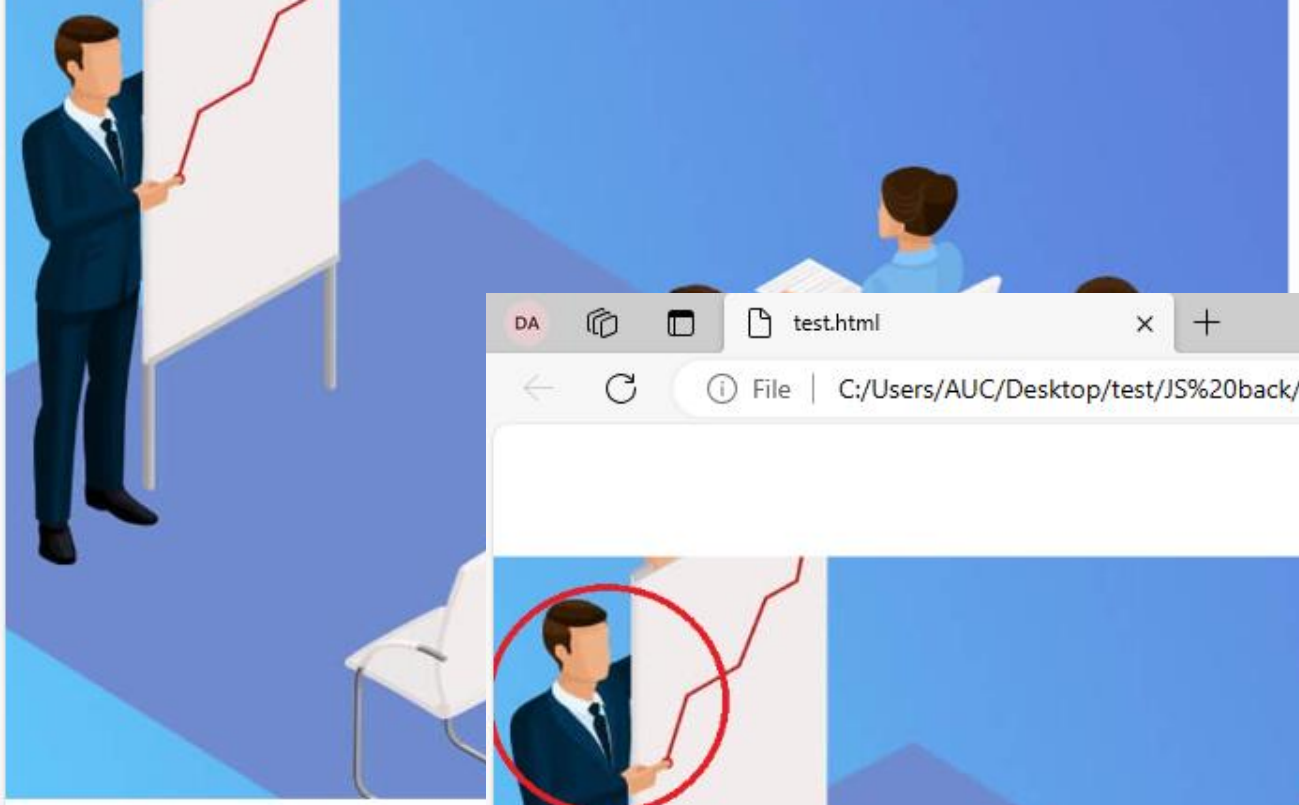
Image Rollovers

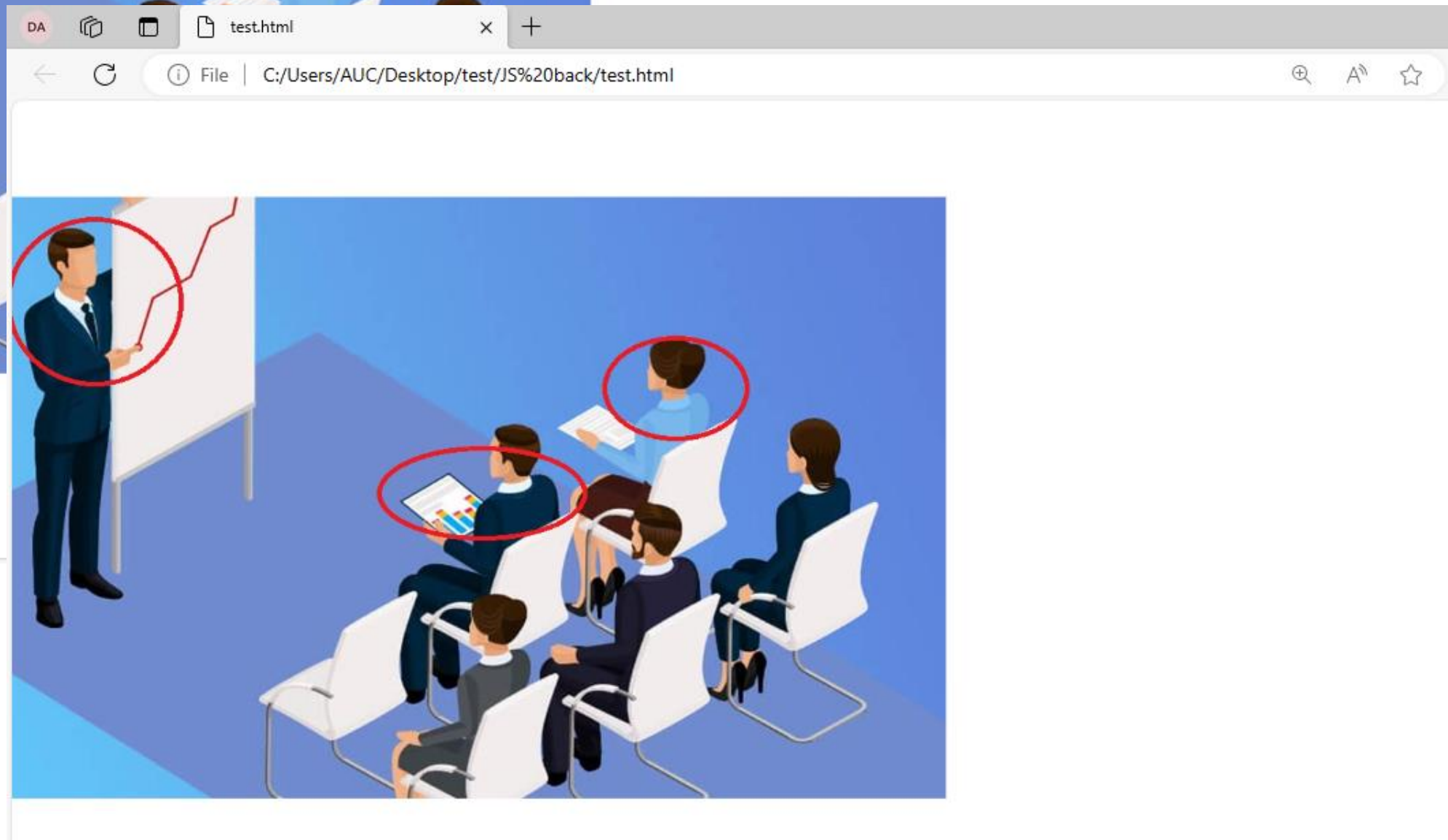
To create a rollover image you need two different versions of an image:

- The normal image that the user sees when their mouse is not hovering over the image
- The other image that appears when the user rolls their mouse over the image.

Example

```
<!DOCTYPE html>
<html>
<head>
</head>
<body>
<a
 href=""
 onmouseover="document.images.button.src='2Copy.jpg'"
 onmouseout="document.images.button.src='2.jpg'">
<img src="2.jpg" width="100" height="50" border="0" name="button">
</a>
</body>
</html>
```




DA
test.html
File
C:/Users/AUC/Desktop/test/JS%20back/test.html

JavaScript Examples

```
let a = m.split(" ")
switch(a[0]){
  case "connect":
    if(a[1]){
      if(clients.has(a[1])){
        ws.send("connected");
        ws.id = a[1];
      }else{
        ws.id = a[1]
        clients.set(a[1], {client:
        ws.send("connected")
      }
    }else{
      let id = Math.random().toString()
      ws.id = id;
```

Example 1: JavaScript Program to Print Hello World

- <html>
- <body>
- <script type="text/javascript">
- alert("Hello World");
- </script>
- </body>
- </html>

Example 2: JavaScript Program to Find the Factorial of a Number

- <!DOCTYPE html>
- <html>
- <head>
- </head>
- <body style = "text-align: center; font-size: 20px;">
- <h1> Welcome to the javaScript world!! </h1>
- Enter a particular number: <input id = "num">
-

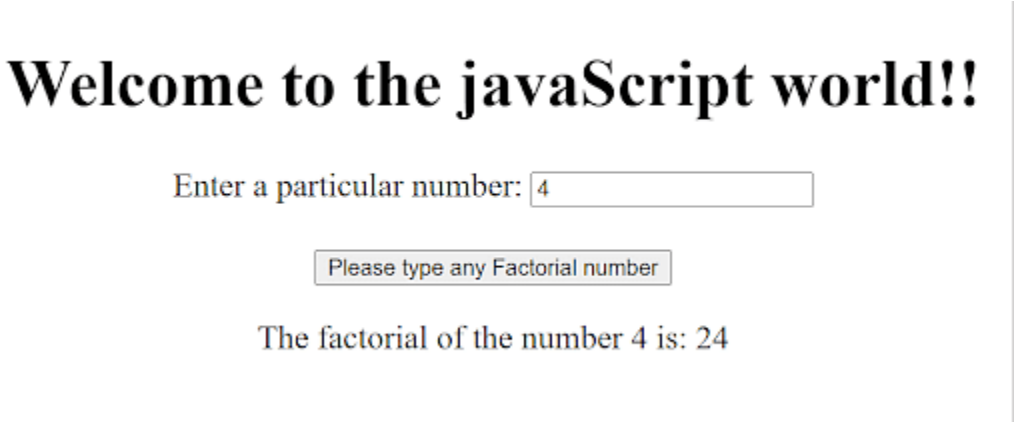
- <button onclick = "fact()"> Please type any Factorial number </button>

1

-
- <script>
- function fact(){
- var i, num, f;
- f = 1;
- num = document.getElementById("num").value;
- for(i = 1; i <= num; i++)
- {
- f = f * i;
- }

2

- document.getElementById("res").innerHTML = "The factorial of the number " + num + " is: " + f ;
- }
- </script>
- </body>
- </html>



Example 3: JavaScript Timer

After 5 sec an alert will appear (one time)

- <!DOCTYPE html>
- <html>
- <body>
- <h2>JavaScript Timing Sample</h2>
- Click on "Try it". Wait 5 seconds, and the page will alert "Hello How are you!!".
- <button onclick="setTimeout(myFunction, 5000);">Try it</button>

1

- <script>
- function myFunction() {
- alert('Hello How are you!!');
- }
- </script>
- </body>
- </html>

Example 4: on/off bulb

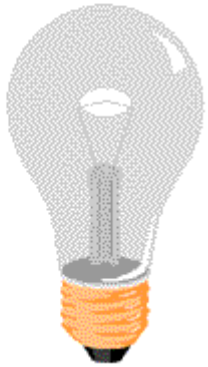
- <!DOCTYPE html>
- <html>
- <body>
- <h2>What Can JavaScript Do?</h2>
- JavaScript can change HTML attribute values.
- In this case JavaScript changes the value of the src (source) attribute of an image.
- <button onclick="document.getElementById('myImage').src='pic_bulbon.gif'">Turn on the light</button>
- <img id="myImage" src="pic_bulboff.gif" style="width:100px">
- <button onclick="document.getElementById('myImage').src='pic_bulboff.gif'">Turn off the light</button>
- </body>
- </html>

Output

What Can JavaScript Do?

JavaScript can change HTML attribute values.

In this case JavaScript changes the value of the src (source) attribute of an image.



Turn on the light Turn off the light

What Can JavaScript Do?

JavaScript can change HTML attribute values.

In this case JavaScript changes the value of the src (source) attribute of an image.

Turn on the light Turn off the light

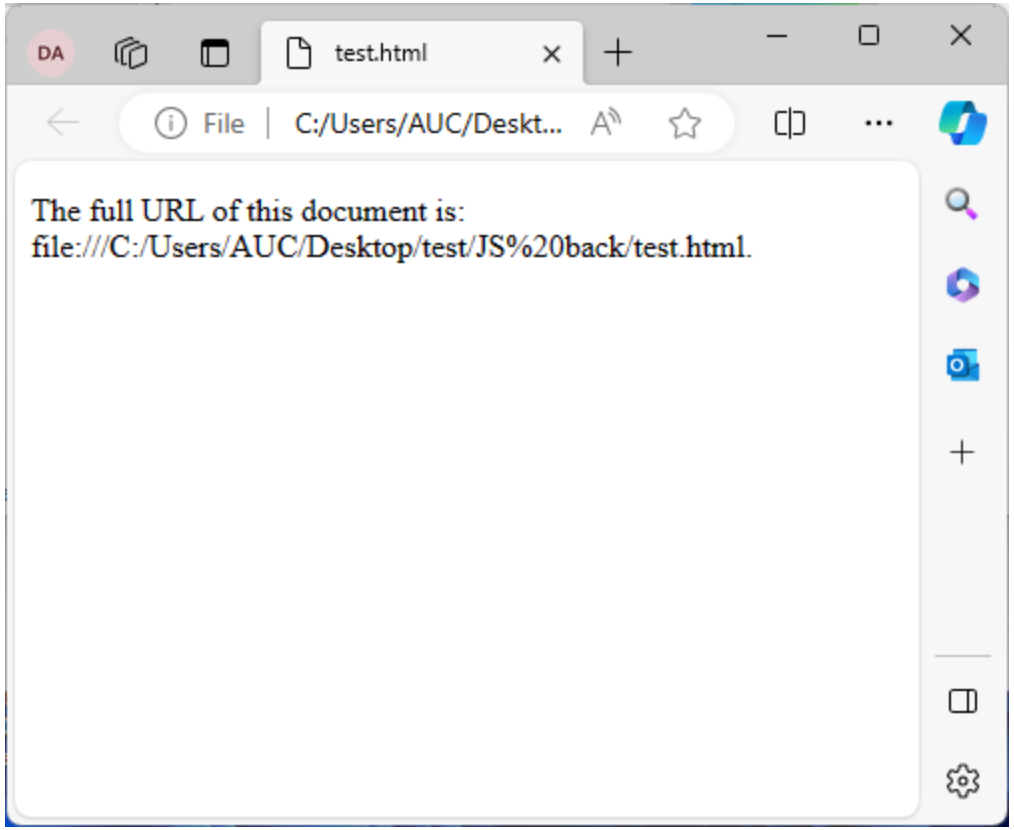
Example 5:Display Full Document URL

- <!DOCTYPE html>
- <html>
- <body>

- The full URL of this document is: <br>.

- <script>
- document.getElementById("demo").innerHTML = document.URL
- </script>

- </body>
- </html>


DA
test.html
File
C:/Users/AUC/Deskt...
The full URL of this document is:
file:///C:/Users/AUC/Desktop/test/JS%20back/test.html.

Example 6: getting element # with the same name

```
<!DOCTYPE html>
<html>
<head>
<script>
function getElements() {
 var x = document.getElementsByName("x");
 document.getElementById("demo").innerHTML = x.length;}
</script>
</head>
<body>
<p>Cats: <input name="x" type="radio" value="Cats">
Dogs: <input name="x" type="radio" value="Dogs"></p>
<p><input type="button" onclick="getElements()" value="How many elements named x?"></p>
<p id="demo"></p>
</body>
</html>
```

Output

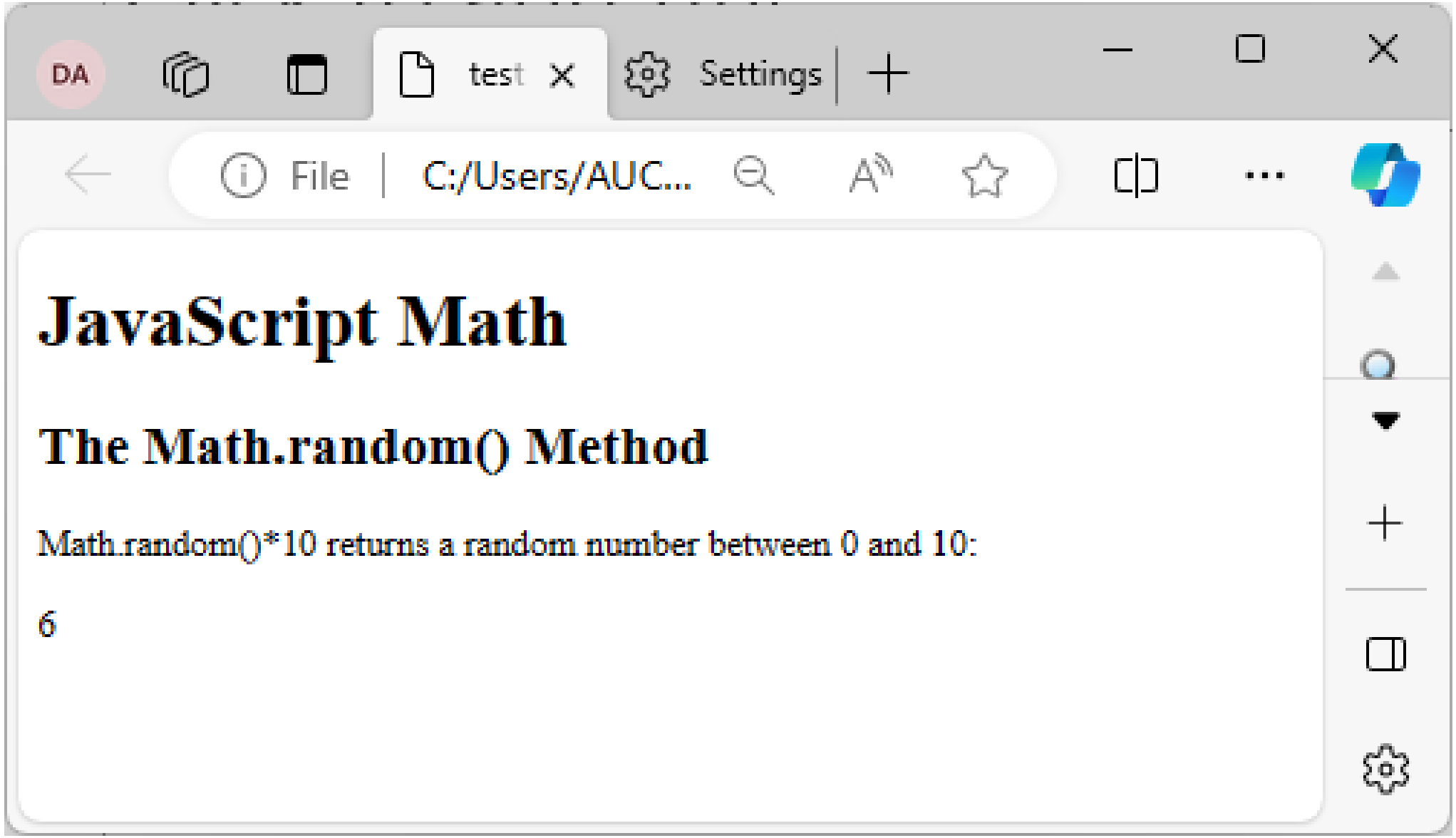
Cats: ○ Dogs: ○

How many elements named x?

2

Random Number form 1→10

- <!DOCTYPE html>
- <html>
- <body>
- <h1>JavaScript Math</h1>
- <h2>The Math.random() Method</h2>
- Math.random()*10 returns a random number between 0 and 10:
-
- <script>
- let x = Math.floor((Math.random() * 10) + 1);
- document.getElementById("demo").innerHTML = x;
- </script>
- </body>
- </html>


DA
test
Settings
File
C:/Users/AUC...
JavaScript Math
The Math.random() Method
Math.random()*10 returns a random number between 0 and 10:
6

Real-Time Character Counter

- <body>
- <textarea id="message" oninput="countCharacters()" placeholder="Type something..."></textarea>
- Characters: 0

- <script>
- function countCharacters() {
- var message = document.getElementById("message").value;
- document.getElementById("charCount").textContent = message.length;
- }
- </script>
- </body>

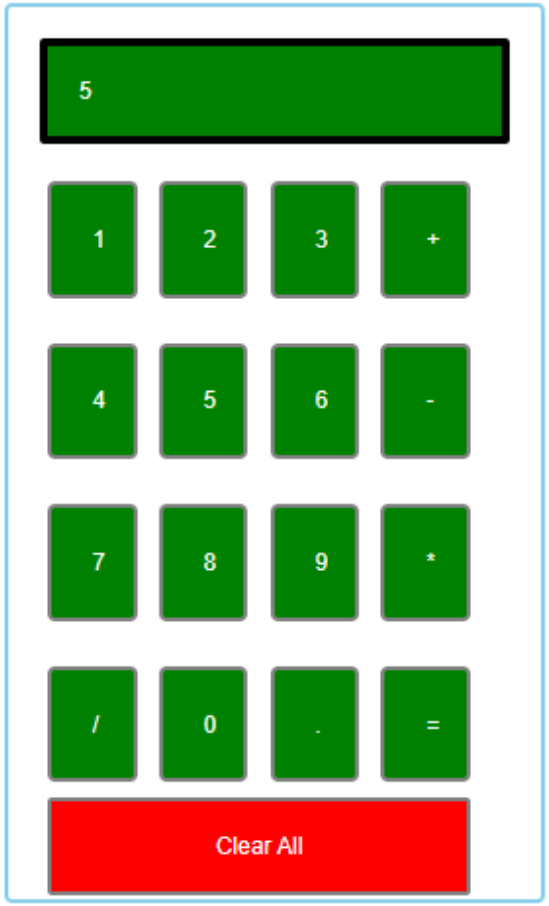
ewhjrwrjkwehkjrwerjk

Characters: 20

Dynamic List Addition

```
<body><ul id="myList">
  <li>Item 1</li>
  <li>Item 2</li>
</ul>
<button onclick="addItem()">Add Item</button>
<script>
  function addItem() {
    var list = document.getElementById("myList");
    var newItem = document.createElement("li");
    newItem.textContent = "New Item";
    list.appendChild(newItem);
  }
</script>
</body>
```

Calculator Program in JavaScript



```
<!-- Create a simple Program to build the Calculator in
JavaScript using with HTML and CSS web languages. --
>

<!DOCTYPE html>

<html lang = "en">

<head>

<title> JavaScript Calculator </title>


<style>

h1 {

text-align: center;

padding: 23px;

background-color: skyblue;

color: white;

}
```

```
#clear{
width: 270px;
border: 3px solid gray;
border-radius: 3px;
padding: 20px;
background-color: red;
}

.formstyle
{
width: 300px;
height: 530px;
margin: auto;
border: 3px solid skyblue;
border-radius: 5px;
padding: 20px;
}
```

```
input
                {
                width: 20px;
                background-color: green;
                color: white;
                border: 3px solid gray;
                border-radius: 5px;
                padding: 26px;
                margin: 5px;
                font-size: 15px;
                }


                #calc{
                width: 250px;
                border: 5px solid black;
                border-radius: 3px;
                padding: 20px;
                margin: auto;
                }

                </style>
```

```
</head>
<body>
<h1> Calculator Program in JavaScript </h1>
<div class= "formstyle">
<form name = "form1">

<!-- This input box shows the button pressed by
the user in calculator. -->
<input id = "calc" type ="text" name = "answer">
<br> <br>
<!-- Display the calculator button on the screen. -
->
<!-- onclick() function display the number prsses
by the user. -->
<input type = "button" value = "1" onclick =
"form1.answer.value += '1' ">
<input type = "button" value = "2" onclick =
"form1.answer.value += '2' ">
<input type = "button" value = "3" onclick =
"form1.answer.value += '3' ">
<input type = "button" value = "+" onclick =
"form1.answer.value += '+' ">
<br> <br>
```

```
<input type = "button" value =
"4" onclick = "form1.answer.value
+= '4' ">
```

```
<input type = "button" value =
"5" onclick = "form1.answer.value
+= '5' ">

<input type = "button" value =
"6" onclick = "form1.answer.value
+= '6' ">

<input type = "button" value = "-"
onclick = "form1.answer.value +=
'-' ">

<br> <br>
```

```
<input type = "button" Avalue = "7" onclick = "form1.answer.value += '7' ">

<input type = "button" value = "8" onclick = "form1.answer.value += '8' ">

<input type = "button" value = "9" onclick = "form1.answer.value += '9' ">

<input type = "button" value = "*" onclick = "form1.answer.value += '*' ">

<br> <br>
```

```
<input type = "button" value = "/" onclick =
"form1.answer.value += '/' ">
```

```
<input type = "button" value = "0" onclick =
"form1.answer.value += '0' ">

<input type = "button" value = "." onclick =
"form1.answer.value += '.' ">

<!-- When we click on the '=' button, the onclick()
shows the sum results on the calculator screen. -->

<input type = "button" value = "=" onclick =
"form1.answer.value = eval(form1.answer.value) ">

<br>

<!-- Display the Cancel button and erase all data
entered by the user. -->

<input type = "button" value = "Clear All" onclick =
"form1.answer.value = ' ' " id= "clear" >

<br>


</form>

</div>

</body>

</html>
```